teleportation—quantum style

The strangest thing about the quantum world is the way an entity such as an electron seems to be in several places at once. In the double-slit experiment, for example, a single electron passing through the apparatus seems to "know" about the entire experimental set-up, and its own place within it. This is called nonlocality, because the electron is not localized at any single point. It is only when the particle interacts with something (such as the TV screen) that the wave function collapses and it becomes localized.

same time, different place
In the quantum world, common-sense ideas about "locality" break down and a particle really can be in more than one place at one time.

the phenomenon of nonlocality

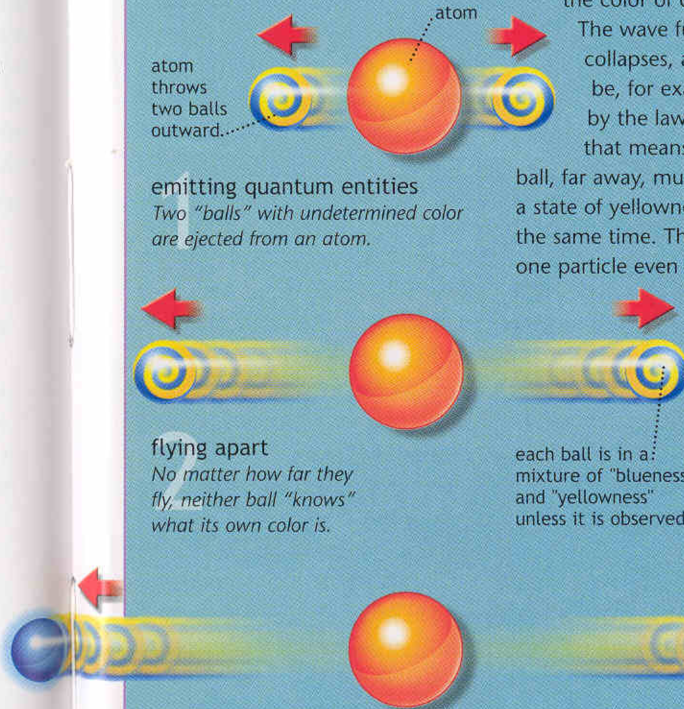
Most electrons are interacting with something all the time. An electron in an atom is interacting with the nucleus of the atom, so it is localized not only to stay in the vicinity of the atom, but at a specific energy level as well. But quantum entities can be prepared in states where they are interacting only in a very limited way with other quantum entities, and then nonlocality shows up with full force.

Experiments to demonstrate quantum nonlocality were devised by the Irish physicist, John Bell, working at CERN in Geneva in the mid-1960s. Several experimental teams took up the challenge of turning these ideas into reality, and the definitive demonstration was carried out by a group working in Paris in the early 1980s. In these

collapsing colors

Quantum nonlocality can be described metaphorically in terms of colored balls. Imagine that balls come in only two colors, blue or yellow. Because of the "law of equal color" an atom can only eject two balls at a time, and there must be one of each color.

But quantum physics says that the color of a ball is not decided until it is measured. The two balls flying off in opposite directions exist in a superposition of states, blue and yellow mixed together to make green, until the measurement is made. Now suppose you measure the color of one of the balls. The wave function collapses, and it is seen to be, for example, blue. But by the law of equal color, that means that the other ball, far away, must collapse into a state of yellowness, at exactly the same time. The balls act like one particle even though they are not in the same location.



emitting quantum entities
Two "balls" with undetermined color are ejected from an atom.

flying apart
No matter how far they fly, neither ball "knows" what its own color is.

long distance effect
At the instant one ball is seen to be yellow, the other becomes blue, without anyone looking at it.